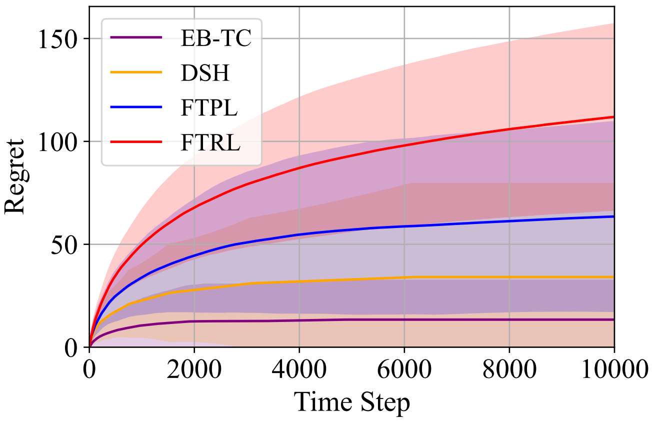

Figure 1: $\mu_1 = [0.55, 0.6, 0.45, 0.3, 0.2]$, where $\Delta_{\min} = 0.05$

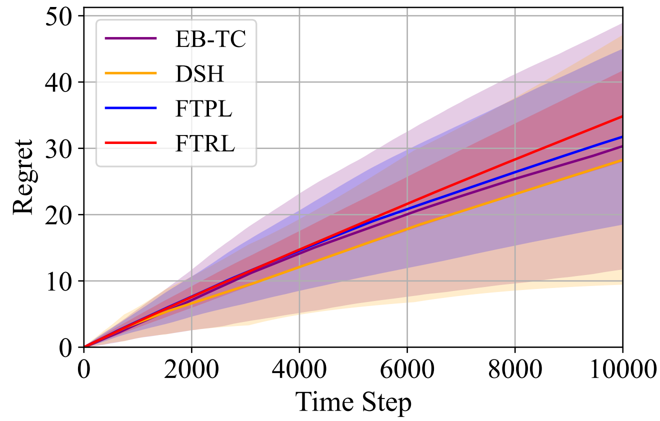

Figure 2: $\mu_2 = [0.905, 0.91, 0.89, 0.908, 0.88]$, where $\Delta_{\min} = 0.002$

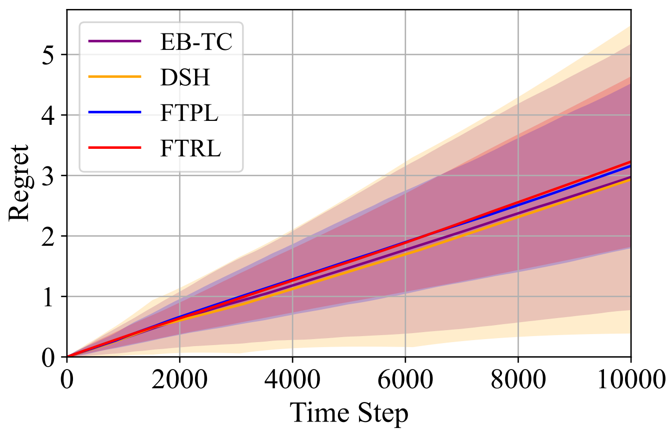

Figure 3: $\mu_3 = [0.201, 0.2008, 0.2007, 0.2009, 0.2]$, where $\Delta_{\min} = 0.0001$

We additionally evaluated the Doubling Sequential Halving (DSH) policy [1], as the reviewer VjFH suggested, in the stochastic setting. While the original DSH only uses the observations for the specific budget period, we utilizes all the observations obtained so far since it outperforms the original DSH. We compare the cumulative regret of FTRL, FTPL, EB-TC, and DSH, considering the five-armed bandit with a unique optimal arm under three different true mean reward configurations. All experiments are conducted for 100 independent repetitions, with the time horizon $T = 10000$.

In Figure 1, one can see that the (stochastic) regrets of the policies converge, which aligns with the theoretical analysis. However, we observe that in this stochastic regime, as the instances become more challenging from Figure 1 to 3, the performance gap between the algorithms gradually narrows, where one can also see that we may require more rounds to converge in such difficult problems. In stochastic setting, EB-TC and DSH outperform both the BOBW policies, FTRL and FTPL. One possible reason for this behavior is on the objective of the policies, where FTPL and FTRL are designed to guarantee BOBW, which requires robust performance across both adversarial and stochastic environments. As a result, it may conservatively select suboptimal arms in certain rounds to remain safe under possible adversarial scenarios, which makes the convergence speed of stochastic regret slow. In contrast, EB-TC and DSH are optimized specifically for stochastic regimes, which makes them to converge faster.

This phenomenon is consistent with prior observations in the BOBW literature. For instance, Zimmert & Seldin (2019) showed that Tsallis-INF, one of the well-known BOBW policies, performed worse than stochastic-specific policies such as Thompson Sampling when evaluated purely in stochastic environments. Similarly, while EB-TC and DSH excel in stochastic settings, our FTPL maintains competitive performance across both stochastic and adversarial regimes, reflecting its intended BOBW design.

References

[1] Yao Zhao, Connor James Stephens, Csaba Szepesvári, and Kwang-Sung Jun. Revisiting simple regret: fast rates for returning a good arm. In *International Conference on Machine Learning*, 2023.